

UNIT 03

THEORIES OF COVALENT BONDS



VALENCE BOND THEORY

POSTULATES

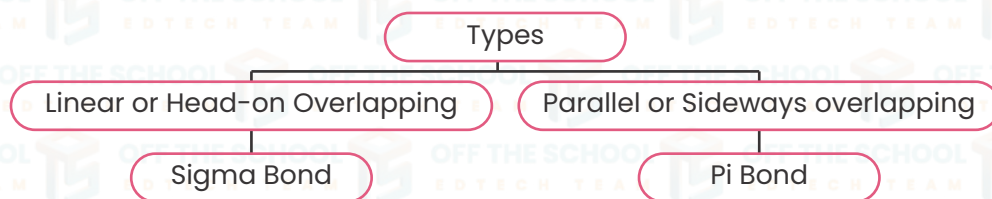
A covalent bond is formed by the overlapping of half-filled orbitals of combining atoms.

1. As a result of overlapping, there is a maximum electron cloud somewhere between two overlapped orbitals.
2. The electrons present in both orbitals should be in opposite spin.
3. The strength of bond is determined by the extent of overlap. The greater the extent overlap the greater will be the strength of bond.

$$\sigma_{s-s} \text{ Overlapping} < \sigma_{s-p} \text{ sp Overlapping} < \sigma_{p-p} \text{ Overlapping}$$

1.0 1.73 3.0

4. Each atom involved in the overlapping keeps its atomic orbital but the electron pair is shared by both atoms that take part in an overlapping.



TYPES OF OVERLAPPING OF ATOMIC ORBITALS

SIGMA BOND (σ)

Definition:

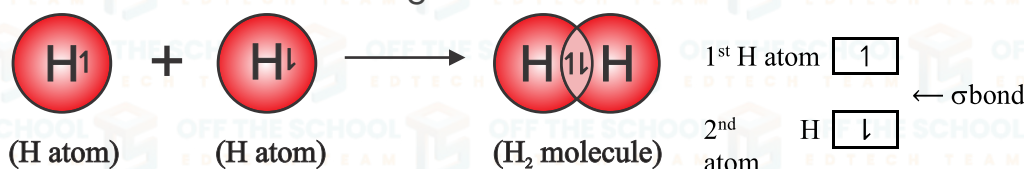
The bond is formed by the linear (head to head) combination or overlapping of two half-filled atomic orbitals.

CHARACTERISTICS:

1. It is formed by head-to-head overlapping of s-s, s-p & p-p orbitals.
2. The electron density in sigma-bonded atoms was found highest between two bonded nuclei.
3. It is a strong bond between half-filled atomic orbitals.
4. Maximum overlapping of orbitals takes place.
5. The free rotation of atoms about sigma bond is possible.
6. Only one lobe of p-orbital takes part in bond formation.

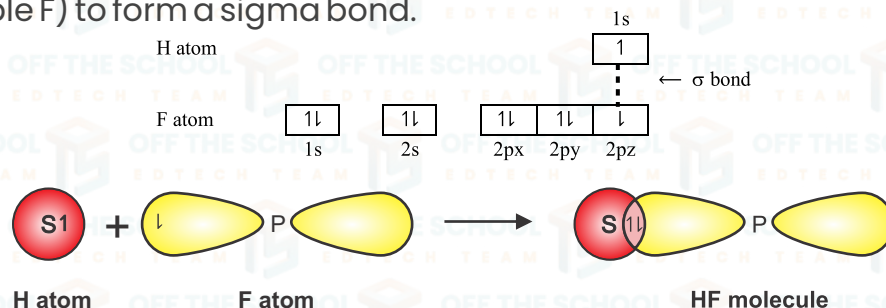
s-s Orbital Overlapping:

- This type of overlapping is found in H_2 molecules.
- When the 1s orbital of one H atom overlaps with the 1s orbital of another H atom to form a sigma bond.
- The electron cloud is rich along the axis between two nuclei.



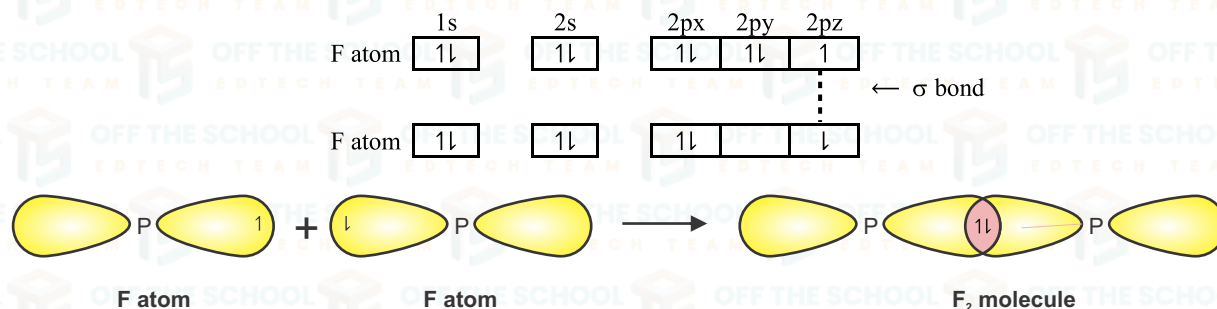
s-p Orbital Overlapping:

- This type of overlapping is found in HF, HCl, and H_2S molecules.
- When the 1s orbital of one H atom overlaps with the 2p orbital of another atom (example F) to form a sigma bond.



p-p Orbital Overlapping:

- This type of overlapping is found in F_2 , Cl_2 , Br_2 , I_2 molecule.
- For example, in F_2 , when the 2p orbital of one F atom overlaps with the 2p orbital of another F atom to form a sigma bond.

**PI BOND (π):****Definition:**

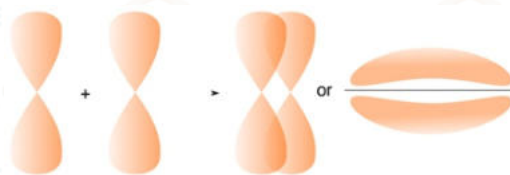
The bond is formed by the lateral (sideways) combination or overlapping of two half-filled atomic orbitals already bonded through the sigma bond.

CONDITION:

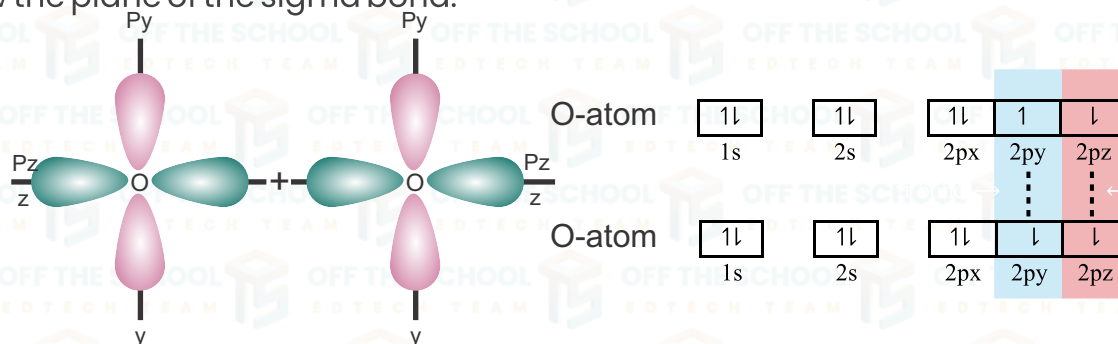
Two p orbitals forming a pi-bond should be parallel to each other.

CHARACTERISTICS:

1. It is formed by the sideways overlapping of p-p Orbitals.
2. It is the weak bond between atomic orbitals.
3. Minimum overlapping of orbitals takes place.
4. The free rotation of atoms about the pi bond is restricted.
5. Both lobes of the p-orbital take part in bond formation.

**Parallel Overlapping of p-p Orbitals:**

- This type of overlapping is found in O_2 & N_2 molecules.
- For Example; in O_2 , Each O atom consists of two partially filled 2px & 2py orbitals.
- 2px orbitals of two O atoms overlap end to end to form a sigma bond.
- 2py orbitals of two O atoms overlap sideways to form a pi bond.
- The electronic cloud of the Pi bond is unsymmetrical & concentrated above & below the plane of the sigma bond.

**STRENGTH OF BOND IN TERMS OF VBT:**

The related strength of a bond is related to the extent of overlapping of the half-filled atomic orbitals.

The relative bond strength is of order, $\pi p-p < \sigma s-s < \sigma s-p < \sigma p-p$

EXPLANATION:

- The greater the overlapping of orbitals, the greater will be bond strength & higher energy is required to break the bond.
- Due to spherical charge distribution in s-orbitals, s-s overlapping is not effective.

- Due to directional charge distribution & longer lobes, p-p linear overlapping is the most effective.
- Therefore, sigma p-p bond is stronger than the sigma s-p bond & sigma s-s bond is relatively weak.
- In pi-bond, the electron cloud density is distributed above & below the plane of the nuclear axis.
- Pi bonds are weaker than sigma bonds.

DIFFERENCE:

SIGMA BOND	PI BOND
It is formed by head-to-head overlapping of atomic orbitals.	It is formed by lateral/parallel/sideways overlapping of atomic orbitals.
Only one sigma bond can be formed between two atoms.	Maximum two pi bonds can be formed between two atoms
The free rotation of atoms about the sigma bond is possible.	The free rotation of atoms about the pi bond is not possible.
The electron cloud is denser at the plane of the bond axis.	The electron cloud is denser above & below the plane of the bond axis.
It is a strong bond between atomic orbitals.	It is the weak bond between atomic orbitals.
Maximum overlapping of orbitals takes place.	Minimum overlapping of orbitals takes place.
Only one lobe of the p-orbital takes part in bond formation	Both lobes of the p-orbital take part in bond formation.

LIMITATIONS OF VBT

- It fails to describe the valences of carbon, boron & beryllium.
- It does not explain the paramagnetic behavior of oxygen molecule.
- It fails to describe the delocalization of electrons in a molecule.

REASON:

Why Pi-bonds are weaker than sigma-bonds?

Pi-bonds are formed by sideways overlapping while sigma bond is formed by end-to-end overlapping. The pi bonds are weaker than sigma bonds as sideways overlapping is less effective as compared to end-to-end overlapping.

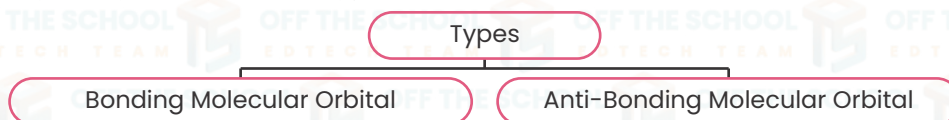
MOLECULAR ORBITAL THEORY**POSTULATES:**

1. Atomic orbitals of similar energies linearly combine to produce molecular orbitals.
2. The combination of two atomic orbitals (AO) gives two molecular orbitals (MO) one has higher energy & other has lower energy than the atomic orbitals from which they are formed.
3. The molecular orbital which has low energy as compared to the energy of atomic orbitals from which they are formed is called as bonding molecular orbital (BMO).
4. The molecular orbital which has high energy as compared to the energy of atomic orbitals from which they are formed is called as anti-bonding molecular orbital (ABMO).
5. The movement of electrons in bonding orbitals is influenced by both nuclei of combining atoms.



6. The filling of electrons into molecular orbital is followed by the fundamental rules of electronic configuration (Aufbau principle, Hund's rule & Pauli rule).

TYPES OF MOLECULAR ORBITALS:



BONDING MOLECULAR ORBITAL

Definition:

The molecular Orbital which has low energy as compared to the energy of atomic Orbitals from which they are formed is called as bonding molecular Orbital (BMO).

CHARACTERISTICS:

1. Electrons present in BMO favor bonding.
2. It is a stable bond between two atomic nuclei.
3. It is formed when lobes of atomic orbitals have the same sign and overlap.
4. The bond is formed between two nuclei and retains its shape.
5. Electrons present in B.M.O's contribute towards attraction between two nuclei.
6. The probability of electron density is maximum between two nuclei.

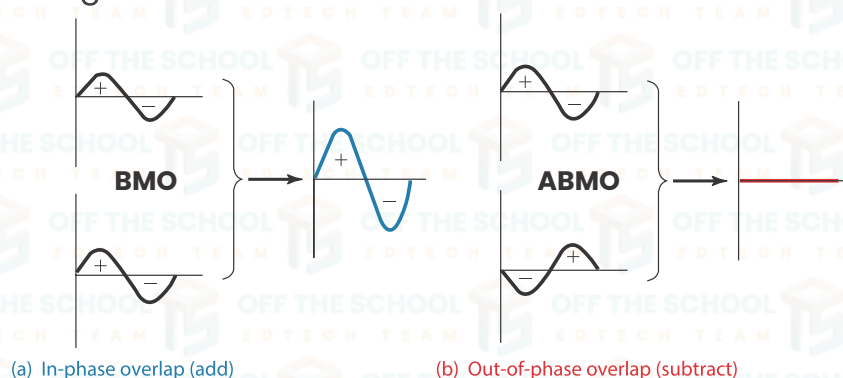
ANTI-BONDING MOLECULAR ORBITAL

Definition:

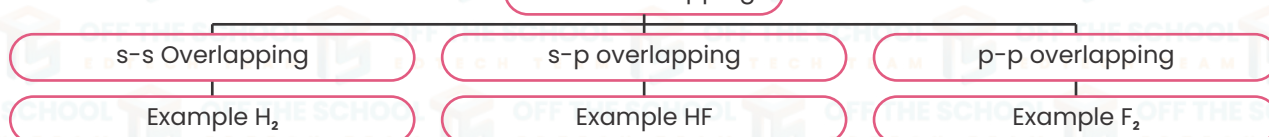
The molecular orbital which has high energy as compared to the energy of atomic orbitals from which they are formed is called as anti-bonding molecular orbital (ABMO).

CHARACTERISTICS:

1. Electrons present in ABMO do not favor bonding.
2. It is the unstable bond between two atomic nuclei.
3. It is formed when lobes of atomic orbitals have different sign overlaps.
4. The bond is formed between two nuclei and they deteriorate.
5. Electrons present in A.B.M.Os contribute towards repulsion between two nuclei.
6. The probability of electron density is zero between two nuclei, as the nodal plane is bisecting the inter-nuclear axis.

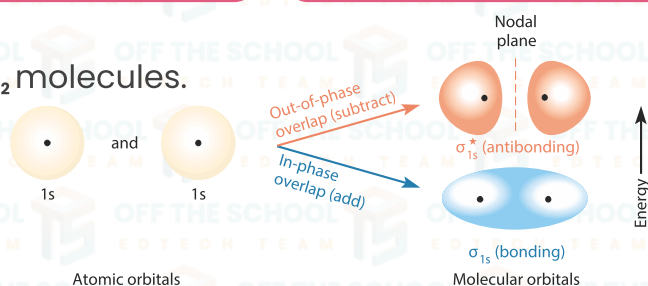


Linear overlapping



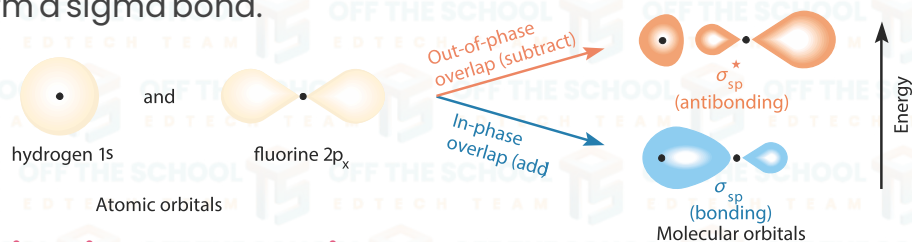
s-s Orbital Overlapping:

- This type of overlapping is found in H_2 molecules.
- When the $1s$ orbital of one H atom overlaps with the $1s$ orbital of another H atom to form a sigma bond.

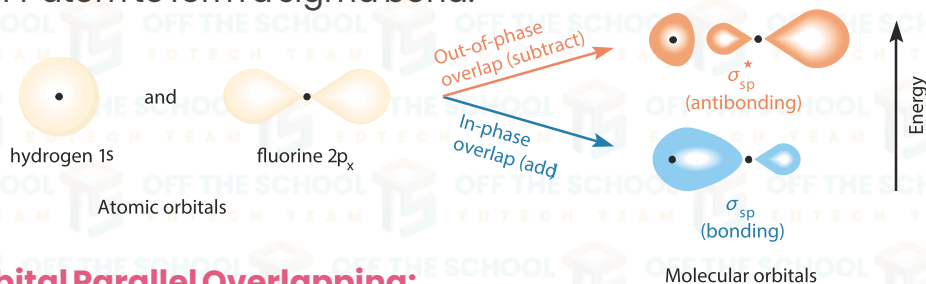


s-p Orbital Overlapping:

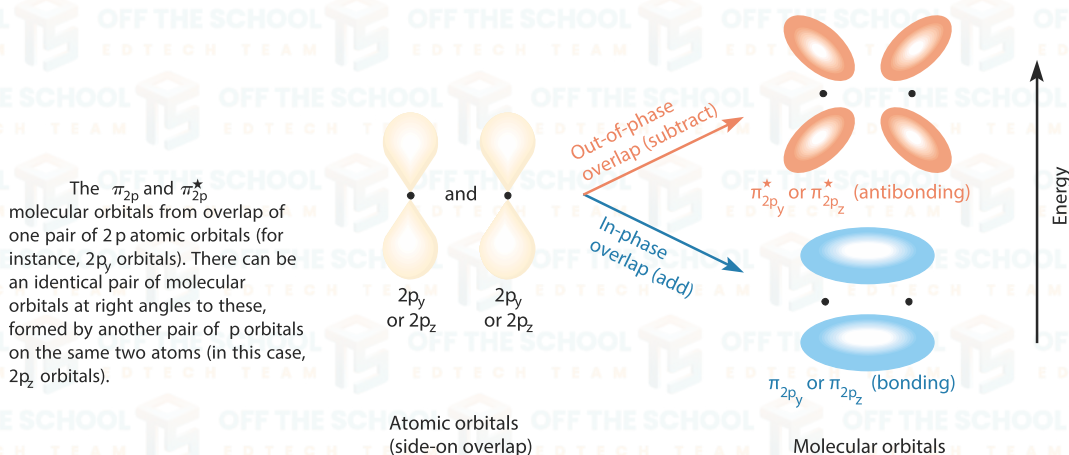
- This type of overlapping is found in HF, HCl, and H₂S molecules.
- When 1s Orbital of one H atom overlaps with 2p orbital of another atom (example F) to form a sigma bond.

**p-p Orbital Linear Overlapping:**

- This type of overlapping is found in F₂, Cl₂, Br₂, and I₂ molecules.
- For example, in F₂, when 2p Orbital of one F atom overlaps with 2p orbital of another F atom to form a sigma bond.

**p-p Orbital Parallel Overlapping:**

- This type of overlapping is found in O₂, and N₂ Molecules
- 2p_y Orbital Of two O atoms overlapping side ways to form π .

**IMPORTANCE OF MOT**

- It is used to predict the bond order.

$$\text{bond order} = \frac{\text{number of electrons in bonding molecular orbitals} - \text{number of electrons in antibonding molecular orbitals}}{2}$$

Where,

Na is no of electrons present in anti-bonding molecular orbital.

Nb is no of electrons present in bonding molecular orbital.

- It is used to predict the magnetic properties of molecules.

MOLECULAR ORBITAL DIAGRAM:**Hydrogen Molecule (H₂)**

Hydrogen has 1 electron in 1s orbital

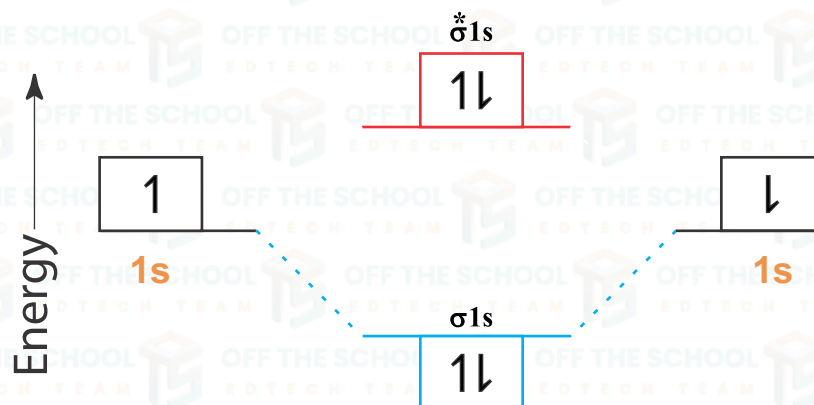
- Hydrogen linearly combines to form two molecular orbitals.
- According to the rule of electronic configuration, all electrons are accommodated in BMO (except n + l rule).
- ABMO remains vacant.
- Bond order can be determined as;



• Bond Order = $\frac{2 - 0}{2} = 1$

- Since there is no unpaired electron present in any molecular orbital. Therefore, hydrogen is diamagnetic.

The energy level diagram of H_2 molecule is expressed as:



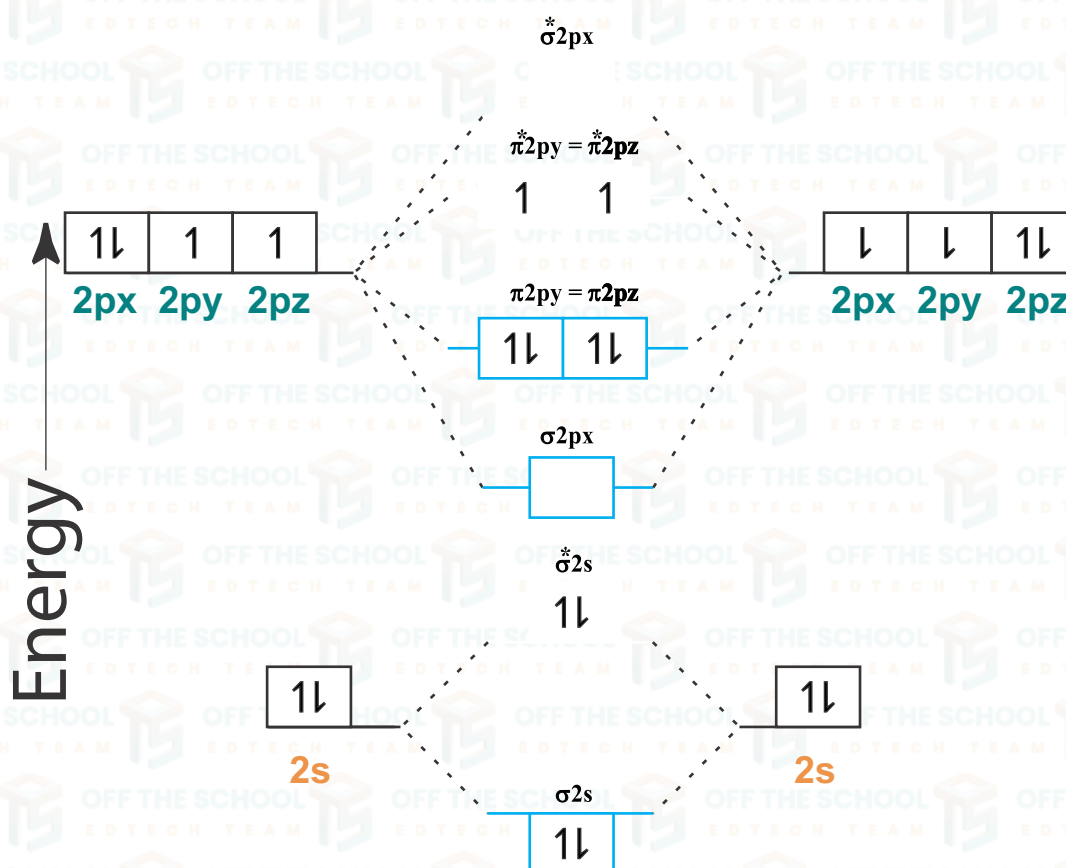
Oxygen Molecule (O_2):

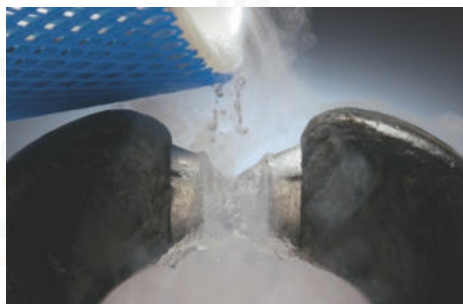
- Oxygen has 08 electrons, which are present in its five orbitals. $1s^2, 2s^2, 2p^4$.
- Five atomic orbitals of both Oxygen atoms combine to form ten molecular orbitals
- According to the rule of electronic configuration, 8 electrons are accommodated in the BMO of the valence shell.
- ABMO of valence shell contains 4 electrons
 $\sigma 1s^2, \sigma 1s^{*2}, \sigma 2s^2, \sigma 2s^{*2}, (\pi 2p_x = \pi 2p_y)^2 \sigma 2p_z^2 (\pi^{*} 2p_x = \pi^{*} 2p_y)^1 \sigma 2p_z^0$
- Bond order can be determined as:

• Bond Order = $\frac{8 - 4}{2} = 2$

- Since there are two unpaired electrons present in the anti-bonding molecular orbital. Therefore, Oxygen is paramagnetic.

The molecular orbital diagram of O_2 is expressed as:





Liquid oxygen is attracted to the poles of a magnet because O_2 is paramagnetic.



Liquid nitrogen is not attracted to the poles of a magnet because N_2 is diamagnetic.

Bonding Molecular Orbitals	Anti-Bonding Molecular Orbitals
1. Molecular orbitals formed by the additive effect (constructive interference) of the atomic orbitals are called bonding molecular orbitals.	Molecular orbitals formed by the subtractive effect (destructive interference) of atomic are called anti-bonding molecular orbitals.
2. The probability of finding the electrons is more in the case of bonding molecular orbitals.	The probability of finding electrons is less in antibonding molecular orbitals. There is also a node between the anti-bonding molecular orbital between two nuclei where the electron density is zero.
3. These are formed by the combination of + with + and - with - part of the electron waves.	These are formed by the overlap of + with - part.
Atomic orbitals: 1s and 1s. In-phase overlap (add) → Molecular orbitals: σ_{1s} (bonding)	Atomic orbitals: 1s and 1s. Out-of-phase overlap (subtract) → Molecular orbitals: σ_{1s}^* (antibonding). Nodal plane is shown between the nuclei.
4. The electron density in the bonding molecular orbital in the internuclear region is high.	The electron density in the antibonding molecular orbital in the internuclear region is very low, so the nuclei are directly exposed to each other.
5. The bonding molecular orbitals are represented by σ , π , δ .	The corresponding anti-bonding molecular orbitals are represented by σ^* , π^* , δ^* .

VALENCE SHELL ELECTRON PAIR REPULSION THEORY (VSEPR)

Introduced By: Sidgwick & Powell

POSTULATES:

1. There are two types of electrons present around the central atom.

Bond Pairs: They are formed by the sharing of unpaired electrons of the central atom & surrounding atom • Bond pair electrons are known as bonding electrons.

Lone Pairs: They have paired electrons of the central atom that don't take part in sharing. Lone pair electrons are known as non-bonding electrons. both are considered active sets of electrons.

2. The active pairs of electrons are similar in charge therefore, they repel each other.



3. The force of repulsion between lone pairs & bond pairs is not the same. They are in order as follows;

Lone-Lone Pair Repulsion > Lone-Bond Pair Repulsion > Bond-Bond Pair Repulsion

4. The active pairs of electrons are arranged far apart from each other to achieve minimum repulsion.

5. In the case of the presence of double & triple bonds, Pi electrons are not considered as active sets of electrons.

6. The shape of the molecules depends upon the number of active sets of electrons (bond pair & lone pair) present around the central atom.

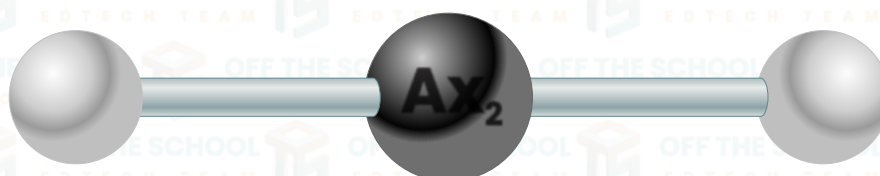
Shapes of the Molecules According To V.S.E.P.R Theory:

Diagonal Shape:

- It is present in AX_2 OR AB_2 -type of molecules.
- When the central atom is surrounded by 2 active sets of electrons & both are bond pairs.
- There is equal repulsion between active sets of electrons & The shape is linear or diagonal.
- The bond angle is 180° .

Example: $BeCl_2$, C_2H_2 , CO_2 .

Molecular Geometry Ball-and-stick models show the shapes of molecules according to VSEPR theory:



Linear

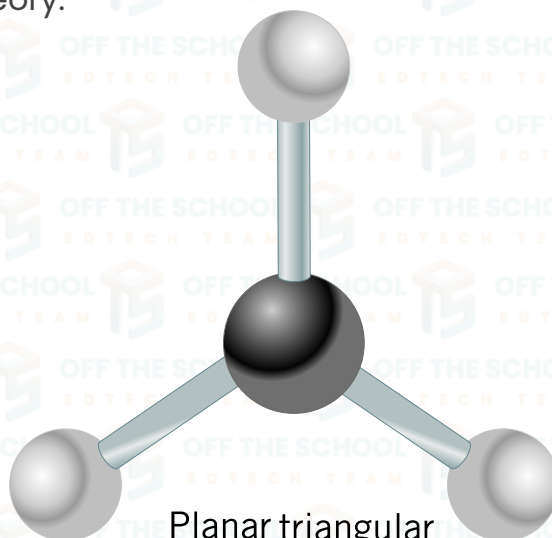
Trigonal Planar Shape:

It is present in AX_3 -type of molecules.

- When the central atom is surrounded by 3 active sets of electrons & all are bond pairs.
- There is equal repulsion between active sets of electrons & The shape is Trigonal planar.
- The bond angle is 120° .

Example: BF_3 , C_2H_4 , $AlCl_3$

Molecular Geometry Ball-and-stick models show the shapes of molecules according to VSEPR theory:

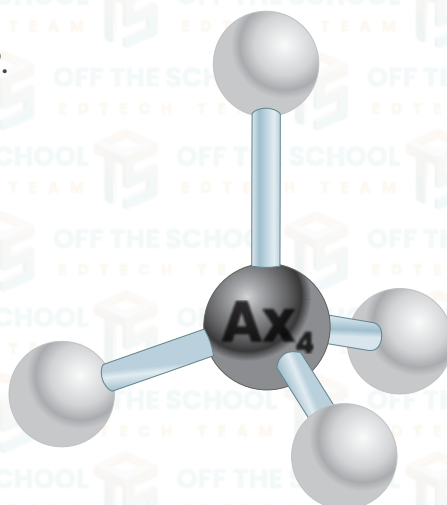


Planar triangular

Tetrahedral Shape:

- It is present in AX_4 -type of molecules.
- When the central atom is surrounded by 4 active sets of electrons & all are bond pairs.
- There is equal repulsion between active sets of electrons & The shape is Tetrahedral.
- The bond angle is 109.28° .

Example: CH_4 , C_2H_6

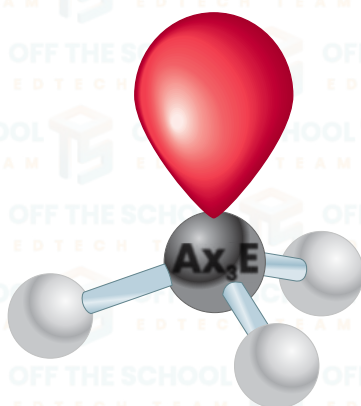


Tetrahedral

Trigonal Pyramidal Shape:

- It is present in AX_3E -type of molecules.
- When the central atom is surrounded by 4 active sets of electrons, 3 are bond pairs & 1 is lone pair.
- There is unequal repulsion between active sets of electrons. The shape is distorted from Tetrahedral to pyramidal.
- The bond angle between the bond pair is reduced from 109.28° to 107° .

Example: NH_3 , PH_3 .

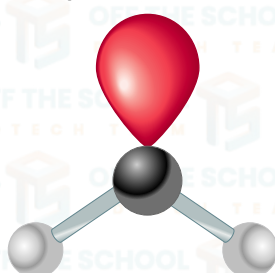


Trigonal Pyramidal

Angular Shape:

- It is present in AX_2E_2 types of molecules.
- When the central atom is surrounded by 4 active sets of electrons, 2 are bond pairs & 2 are lone pairs.
- There is unequal repulsion between active sets of electrons. The shape is distorted from Tetrahedral to angular.
- The bond angle between the bond pair is reduced from 109.28° to 104.5° .

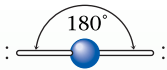
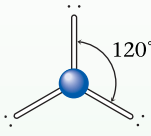
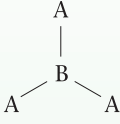
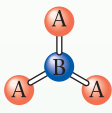
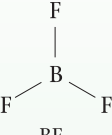
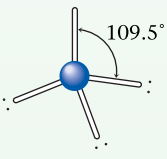
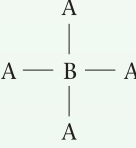
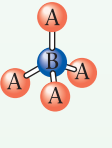
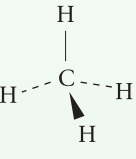
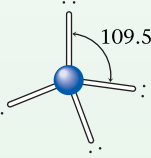
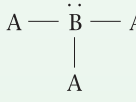
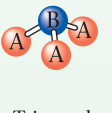
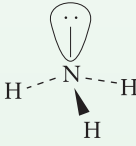
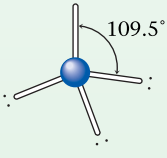
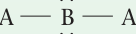
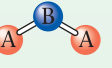
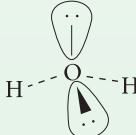
Example: H_2O , H_2S .

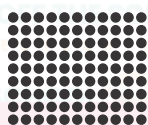


Nonlinear (bent)



Arrangements of Electron Pairs and the Resulting Molecular Structures for Two, Three, and Four Electron Pairs

Case	Number of Electron Pairs	Bonds	Electron Pair Arrangement	Partial Lewis Structure	Molecular Structure	Example
1	2	2	 Linear	A — B — A	 Linear	F — Be — F BeF ₂
2	3	3	 Trigonal planar (triangular)		 Trigonal planar (triangular)	 BF ₃
3	4	4	 Tetrahedral		 Tetrahedral	 CH ₄
4	4	3	 Trigonal pyramidal		 Trigonal pyramid	 NH ₃
5	4	2	 Tetrahedral		 Bent or V-shaped	



Problem-Solving Tip:

Working with MO Theory

MO theory is often the best model to predict the bond order, bond stability, or magnetic properties of a molecule or ion. The procedure is as follows:

1. Draw (or select) the appropriate MO energy level diagram.
2. Count the total number of electrons in the molecule or ion.
3. Follow the Pauli Exclusion Principle and Hund's Rule to add the electrons to the MO diagram.
4. Calculate the bond order: $\text{Bond order} = \frac{(\text{bonding e}^- - \text{antibonding e}^-)}{2}$.
5. Use the bond order to evaluate stability.
6. Look for the presence of unpaired electrons to determine if a species is paramagnetic.



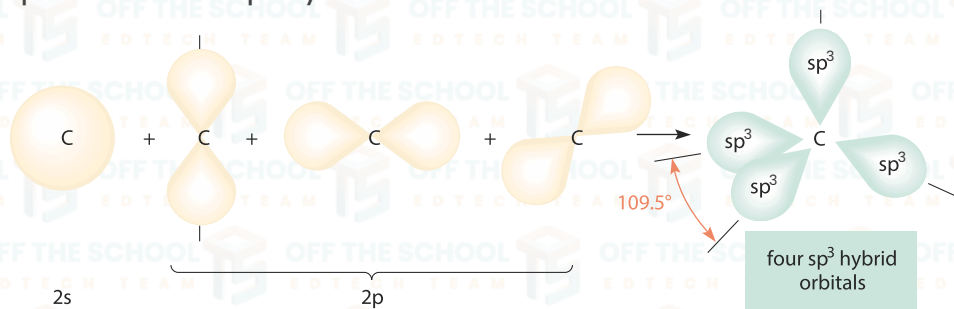
HYBRIDIZATION

Definition:

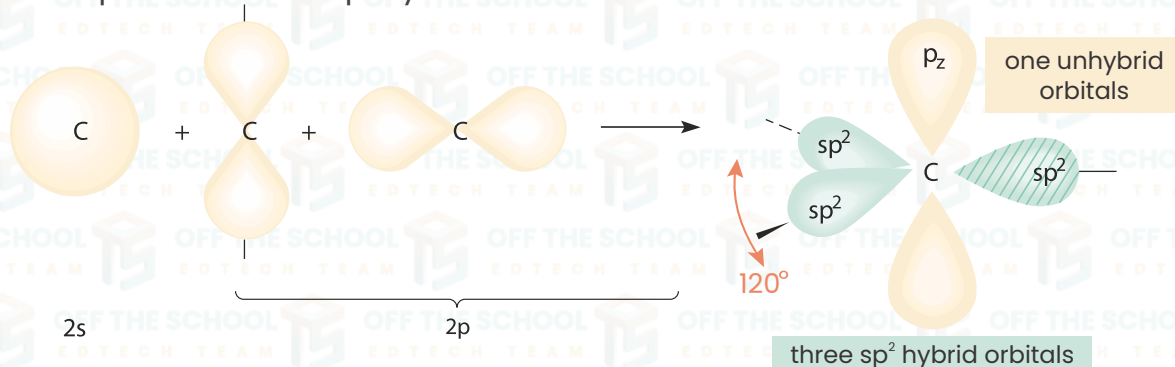
- **Hybridization:** The mixing of different atomic orbitals to produce the same number of equivalent orbitals, having the same shape & energy.
- **Hybrid Orbitals:** Orbitals which are equivalent having the same shape & same energy after hybridization.
- **Un Hybrid Orbitals:** Orbitals which does not take part in hybridization & retain their shape.

Types of Hybridization

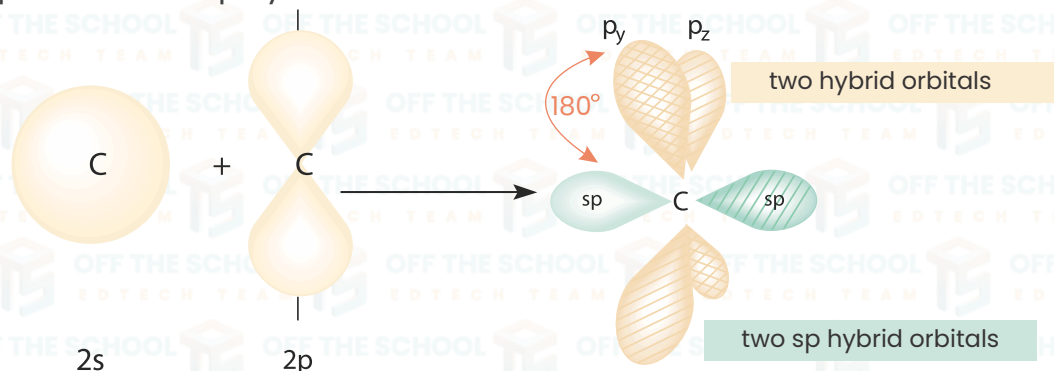
- **sp^3 Hybridization:** The process of combination of one S orbital & three P orbitals, intermix to produce four sp^3 hybrid orbitals.



- **sp^2 Hybridization:** The process of combination of one S orbital & two P orbitals, intermix to produce three sp^2 hybrid orbitals.



- **sp Hybridization:** The process of combination of one S orbital & one P orbital, intermix to produce two sp hybrid orbitals.



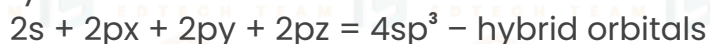
Mixing Orbitals		No & Types of Orbital Produced	Types of Hybridization Types	The ratio of Combination S & P	S Character %	P Character %
1	3	4 sp^3	Sp^3	1:3	25	75
1	2	4 sp^2	Sp^2	1:2	33.3	66.6
1	1	4 sp	S	1:1	50	50

Structure Of Alkane (sp^3 Hybridization in ethane):**Hybridization of Carbon****Ground State****Excited State****Hybridized State**

Total covalent bonds	07
Total sigma bonds	07
Total pi-bonds	00
Type of σ bond	$sp^3 - s$: 06, $sp^3 - sp^3$: 01
Bond angle	109.5°
Bond length	1.54°
Shape	Tetrahedral

sp^3 hybridization: is the type of hybridization in which one s and three p orbitals overlap each other to form four sp^3 hybridized orbitals.

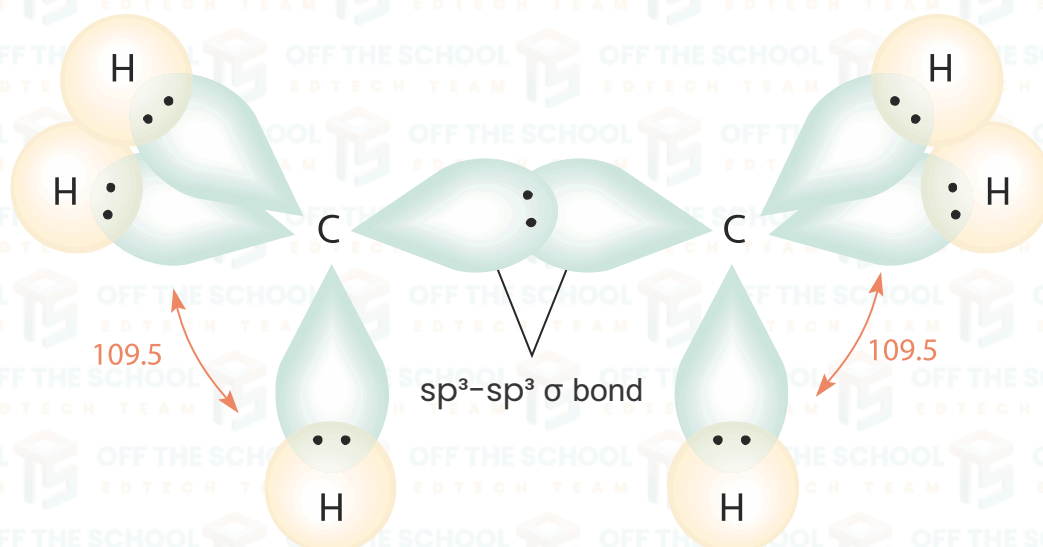
Formation: In sp^3 -hybridization, s & p orbitals intermix with each other in a ratio 1:3 to produce 4 sp^3 -hybrid orbitals.



- **Characteristics:** They possess 25% character of s orbital & 75% character of p orbitals.
- **Geometry:** sp^3 orbitals are directed towards the four corners of the regular tetrahedron in which each angle is 109.28° .
- **Occurrence:** sp^3 -hybridization occurs in saturated hydrocarbons in which carbon is attached with four hydrogen atoms.

 sp^3 hybridization in ethane:

- Ethane is composed of two carbon atoms and six hydrogen atoms, giving it the molecular formula C_2H_6 .
- In the formation of bonds, the 2s orbital and three 2p orbitals of each carbon atom mix together to form four sp^3 hybridized orbitals.
- The resulting geometry of ethane is tetrahedral, with bond angles of 109.5° .
- Each sp^3 hybrid orbital on the carbon atom overlaps with an s orbital of a hydrogen atom, and the sp^3 hybrid orbital of the other carbon atom to form six sigma (σ) bonds in total.



Structure Of Alkene (sp^2 Hybridization in ethene):

Electronic configuration of C-atom

Ground State



Excited State



Hybridized State



Total covalent bonds:	06
Total sigma bonds	05
Total pi-bonds;	01
Type of σ bond:	$sp^2 - s$: 04, $sp^2 - sp^2$: 01
Type of π bond:	$P_z - P_z$
Bond angle:	120°
Bond length:	1.34 $\text{\AA}$
Shape:	Trigonal planar

sp^2 Hybridization: is the type of hybridization in which one s orbital and two p orbitals overlap each other to form three sp^2 hybridized orbitals.

Formation: In sp^2 -hybridization, s & p orbitals intermix with each other in a ratio 1:2 to produce 3 sp^2 hybrid orbitals. One P Orbital (2p orbital) which is not involved in hybridization remains un-hybridized and is present right angle to the plane of sp^2 -hybrid orbitals.



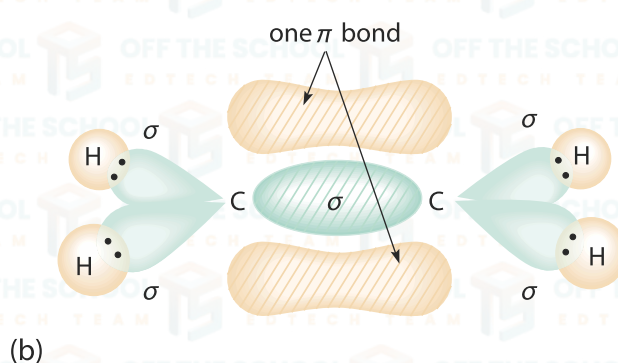
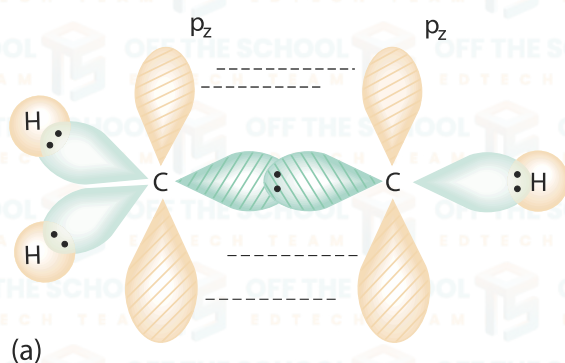
Characteristics: They possess 33.3% character of s orbital & 66.7% character of p orbitals.

Geometry: sp^2 orbitals are coplanar & directed towards the three corners of an equilateral triangle in which each angle is 120° .

Occurrence: sp^2 -hybridization occurs in unsaturated hydrocarbons in which carbon is attached with three atoms or a group of atoms.

sp^2 Hybridization in Ethene:

- Ethene consists of two carbon atoms and four hydrogen atoms; therefore, the molecular formula of ethene is C_2H_4 .
- Before the formation of bonds, one 2s orbital and two 2p orbitals of each carbon atom are mixed together to form three sp^2 hybrid orbitals, while the remaining $2p_z$ orbital of each carbon atom remains unhybridized and lies perpendicular to the plane of the sp^2 orbitals.
- Two sp^2 hybrid orbitals of each carbon atom overlap with the s orbital of two hydrogen atoms forming two sigma bonds (sp^2-s), while the third sp^2 hybrid orbital overlaps with the sp^2 orbital on the other carbon atom forming another sigma bond (sp^2-sp^2). Therefore, there are overall four sigma bonds formed in each molecule.



The unhybridized $2p_z$ orbitals of each carbon atom are parallel to each other and perpendicular to the plane of the sp^2 hybrid orbitals. They overlap side by side (laterally) to form a pi bond ($p-p$).

Sp Hybridization: sp hybridization is the type of hybridization in which one s orbital and one p orbital overlap each other to form two sp hybrid orbitals.

Structure of Alkyne: (sp Hybridization in Ethyne)

Electronic Configuration of C-atom

Ground State



Excited State



Hybridized State



Total covalent bonds:	05
Total sigma bonds:	03
Total pi-bonds:	02
Type of σ bond:	$Sp-s$: 02, $Sp-sp$: 01
Type of π bond:	P_y-P_y & P_z-P_z
Bond angle:	180°
Bond length:	1.20°
Shape:	Linear

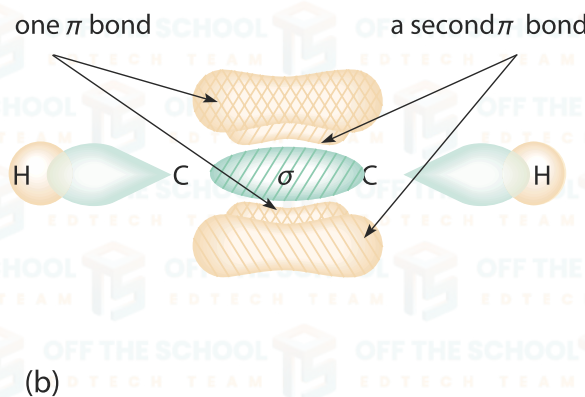
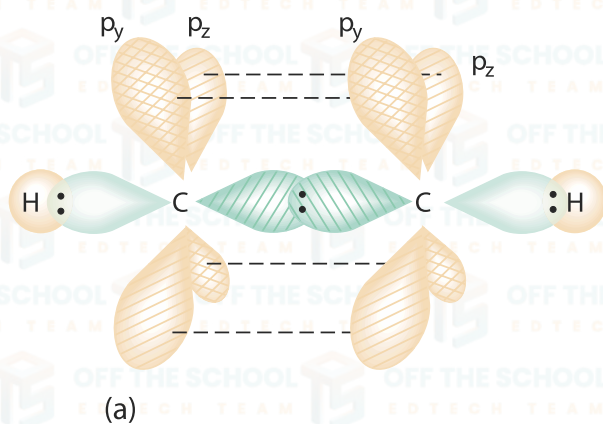
Ethyne consists of two carbon atoms and two hydrogen atoms; therefore, the molecular formula of ethyne is C_2H_2 .

Before the formation of bonds, one $2s$ orbital and one $2p$ orbital of each carbon atom are mixed together to form two sp hybrid orbitals, while the remaining $2p_y$ and $2p_z$ orbitals of each carbon atom stay unhybridized and lie perpendicular to the plane of the sp orbitals.

One sp hybrid orbital of each carbon atom overlaps with the s orbital of a hydrogen atom forming one sigma bond ($sp-s$), while the other sp orbital overlaps with the sp orbital on the other carbon atom forming another sigma bond ($sp-sp$). Therefore, there are two sigma bonds formed in the molecule.

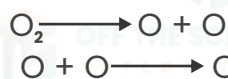
The $2p_y$ and $2p_z$ unhybridized orbitals of each carbon atom are parallel to each other and perpendicular to the line of the sp orbitals. They overlap side by side (laterally) to form two pi bonds ($p-p$).

The geometry of ethyne is linear, and the bond angle between the hydrogen carbon bonds is 180° .



BOND ENERGY**Definition:**

The energy involved for the breaking or formation of 1 mole of the particular type of bonds in a molecule.



$$\Delta H = +495 \text{ kJ/mol}$$

Endothermic

$$\Delta H = -495 \text{ kJ/mol}$$

Exothermic

BOND DISSOCIATION ENERGY:**Definition:**

The energy required for the breaking of 1 mole of the particular type of bonds in a molecule.



$$\Delta H = +495 \text{ kJ/mol}$$

Endothermic

COMPARISON OF BOND ENERGY & BOND STRENGTH:**Polarity:**

- The bond energy of diatomic molecules depends upon polarity.
- Homo diatomic molecules are non-polar while hetero diatomic molecules are polar.
- The bond energy of hetero diatomic polar molecules is relatively greater.



$$\Delta H = +431 \text{ kJ/mol}$$



$$\Delta H = +242 \text{ kJ/mol}$$

Size of bonded atoms:

- Smaller atoms are bonded more strongly due to short inter-nuclear distance.
- Bond energy for the dissociation of smaller bonded atoms is relatively greater.



$$\Delta H = +436 \text{ kJ/mol}$$



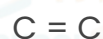
$$\Delta H = +242 \text{ kJ/mol}$$

Presence of multiple bonds:

Molecules with multiple bonds required greater bond energy:



$$\Delta H = +348 \text{ kJ/mol}$$



$$\Delta H = +614 \text{ kJ/mol}$$



$$\Delta H = +839 \text{ kJ/mol}$$

BOND LENGTH:**Definition:**

- The distance between the nuclei of two bonded atoms in a molecule.
- Bond length is experimentally determined by X-ray diffraction technique.
- It is a very small distance measured as Angstrom ($\text{\AA}$) which is equal to 10^{-10} m .

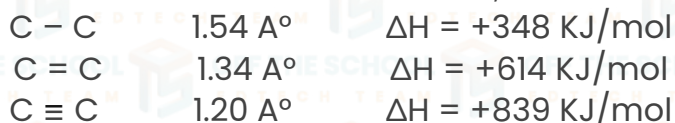


COMPARISON OF BOND LENGTH & BOND STRENGTH:**Polarity:**

- The bond length between two different atoms is shorter than similar atoms.

**Bond Order:**

- Greater the number of bonds between two atoms, the shorter the bond length

**IONIC CHARACTER OF COVALENT BOND:**

- The covalent bond is formed by the mutual sharing of electrons when a covalent bond is formed between two similar atoms they have the same electronegativity thus electrons are equally attracted by both atoms & poles are not observed on atoms.

• However, when a covalent bond is formed between two dissimilar atoms they have different electronegativity due to the difference in electronegativity the shared electrons are not equally attracted by both atoms & electrons are attracted towards more electronegative atoms due to which poles appeared on atoms. The poles are represented as δ^+ & δ^-

- These δ^+ & δ^- are partial electrostatic charges. Thus, a covalent compound is not pure covalent but contains a partial ionic character due to the presence of poles. The whole phenomenon is known as the ionic character of a covalent bond.

EXAMPLE:

HCl, (E.N. H = 2.1, Cl = 3.0 Therefore, $\Delta \text{E.N.} = 0.9$)

COMPARISON OF IONIC CHARACTER & BOND STRENGTH:**Electronegativity Difference:**

- The ionic character of the covalent bond depends upon the electronegativity difference between combining atoms.
- The greater the electronegativity difference between combining atoms, the greater will be the bond strength
- When the electronegativity difference between combining atoms is almost the same, the relative size of atoms decides the bond strength of the ionic character.

Bond	$\Delta \text{E.N.}$	Ionic Character
H - F	1.9	43%
H - Cl	0.9	17%
H - Br	0.7	12%
H - I	0.3	6%

DIPOLE MOMENT:**DEFINITION:**

- The product of the magnitude of charge and the distance between positive & negative centers.

FORMULA:

$$\mu = e \times d$$

Where, "e" is the charge on the pole & "d" is the distance between poles.

EXPLANATION:

- The dipole moment decides the polar character in molecules in the case when



two dissimilar atoms combine to form a covalent bond.

- Polarity exists in angular molecules.
- In linear molecules, the dipole moment of one side cancels the dipole moment of the other side. Hence, the molecule becomes non-polar.
- The net dipole moment is shown by an arrow ($\rightarrow$) & it is a vector quantity.

UNIT:

The SI unit of Dipole moment is Coulomb. Meter (C.m) or Debye.

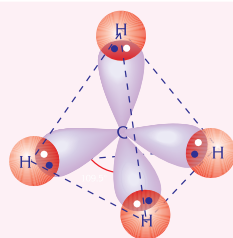
Experimentally charge is in order of 10^{-10} esu & distance between both centers is in the range of 10^{-8} cm.

Therefore, $1 \text{ Debye} = 10^{-10} \text{ esu} \times 10^{-8} \text{ cm}$

$1 \text{ Debye} = 1 \times 10^{-18} \text{ esu.cm}$

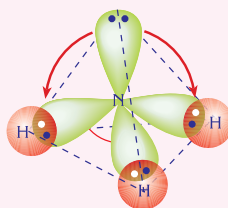
$1 \text{ Debye} = 3.336 \times 10^{-30} \text{ C.m}$

SAME HYBRIDIZATION BUT DIFFERENT ANGLE

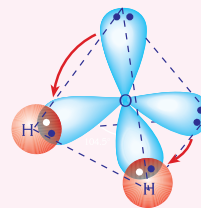


Molecule formula :
Molecule geometry :
Polarity of the molecule :
Bond angle :

CH_4
tetrahedral
nonpolar
 $\text{HCH} : 109.5^\circ$



NH_3
trigonal pyramidal
polar
 $\text{HNH} : 107^\circ$



H_2O
bent
polar
 $\text{HOH} : 104.5^\circ$

The shape of methane (CH_4), ammonia (NH_3) and water (H_2O) molecules are based upon the tetrahedron. CH_4 has four bonds while the other two molecules have lone pair electrons as well as bonding electron pairs. As the number of lone pair electrons increases the bond angles decrease. The reason is that the free electron pairs occupy a larger volume in space. As a result, the shape of CH_4 is tetrahedral, but due to the greater repulsive forces of free electron pairs the shape of NH_3 is trigonal pyramidal with a bond angle of 107° and that of H_2O is bent with a bond angle of 104.5° .

To summarize, the orientations of the electron pairs around of the central atoms in each of the three molecules are based upon the tetrahedron. The shape of methane is tetrahedral, but in the ammonia and water molecules due to the repulsive forces of the non-bonding electrons, the shapes are trigonal pyramidal and bent respectively, with a decreasing bond angle.

